

1.5 THE TRAJECTORY EQUATION

Earlier we wrote the equation of motion for a small mass orbiting a large central body. While this equation (1.3-3) is simple, its complete solution is not. A partial solution which will tell us the size and shape of the orbit is easy to obtain. The more difficult question of how the satellite moves around this orbit as a function of time will be postponed to Chapter 4.

1.5.1 Integration of the Equation of Motion. You recall that the equation of motion for the two-body problem is

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r}.$$

Crossing this equation into $\mathbf{h}$ leads toward a form which can be integrated:

$$\ddot{\mathbf{r}} \times \mathbf{h} = -\frac{\mu}{r^3} \mathbf{r} \times \mathbf{h} \quad (1.5-1)$$

The left side of equation (1.5-1) is clearly $d/dt (\dot{\mathbf{r}} \times \mathbf{h})$ - try it and see. Looking for the right side to also be the time rate of change of some vector quantity, we see that

$$\begin{aligned} \frac{\mu}{r^3} (\mathbf{h} \times \mathbf{r}) &= \frac{\mu}{r^3} \mathbf{h} \times \mathbf{r} \\ &= \frac{\mu}{r} \mathbf{v} - \frac{\mu \dot{r}}{r^2} \mathbf{r} \end{aligned}$$

since $\mathbf{r} \cdot \dot{\mathbf{r}} = r \dot{r}$. Also note that μ times the derivative of the unit vector is also

$$\mu \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right) = \frac{\mu}{r} \mathbf{v} - \frac{\mu \dot{r}}{r^2} \mathbf{r}.$$

We can rewrite equation (1.5-1) as

$$\frac{d}{dt} (\dot{\mathbf{r}} \times \mathbf{h}) = \mu \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right).$$

Integrating both sides

$$\dot{\mathbf{r}} \times \mathbf{h} = \mu \frac{\mathbf{r}}{r} + \mathbf{B}. \quad (1.5-2)$$

Where $\mathbf{B}$ is the vector constant of integration. If we now dot multiply this equation by $\mathbf{r}$ we get a scalar equation:

$$\mathbf{r} \cdot \dot{\mathbf{r}} \times \mathbf{h} = \mathbf{r} \cdot \mu \frac{\mathbf{r}}{r} + \mathbf{r} \cdot \mathbf{B}.$$

Since, in general, $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$ and $\mathbf{a} \cdot \mathbf{a} = a^2$

$$h^2 = \mu r + r B \cos \nu$$

where ν (nu) is the angle between the constant vector $\mathbf{B}$ and the radius vector $\mathbf{r}$. Solving for r , we obtain

$$r = \frac{h^2/\mu}{1 + (B/\mu) \cos \nu}. \quad (1.5-3)$$

1.5.2 The Polar Equation of a Conic Section. Equation (1.5-3) is the trajectory equation expressed in polar coordinates where the polar angle, ν , is measured from the fixed vector $\mathbf{B}$ to $\mathbf{r}$. To determine what kind of a curve it represents we need only compare it to the general equation of a conic section written in polar coordinates with the origin located at a focus and where the polar angle, ν , is the angle between $\mathbf{r}$ and the point on the conic nearest the focus:

$$r = \frac{p}{1 + e \cos \nu} \quad (1.5-4)$$

In this equation, which is mathematically identical in form to the trajectory equation, p is a geometrical constant of the conic called the "parameter" or "semi-latus rectum." The constant e is called the "eccentricity" and it determines the type of conic section represented by equation (1.5-4).

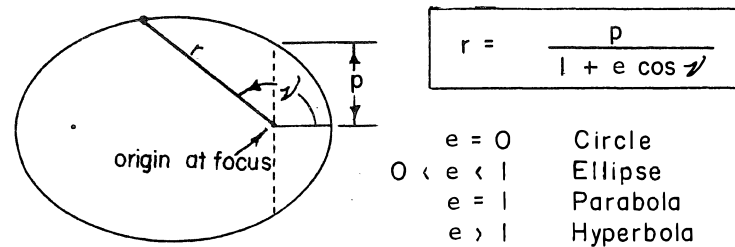


Figure 1.5-1 General equation of any conic section in polar coordinates

The similarity in form between the trajectory equation (1.5-3) and the equation of a conic section (1.5-4) not only verifies Kepler's first law but allows us to extend the law to include orbital motion along any conic section path, not just ellipses.

We can summarize our knowledge concerning orbital motion up to this point as follows:

1. The family of curves called "conic sections" (circle, ellipse, parabola, hyperbola) represent *the only possible paths* for an orbiting object in the two-body problem.
2. The focus of the conic orbit *must* be located at the center of the central body.
3. The mechanical energy of a satellite (which is the sum of kinetic and potential energy) does not change as the satellite moves along its conic orbit. There is, however, an exchange of energy between the two forms, potential and kinetic, which means that the satellite must slow-down as it gains altitude (as r increases) and speed-up as r decreases in such a manner that $\mathcal{E}$ *remains constant*.
4. The orbital motion takes place in a plane which is *fixed in inertial space*.
5. The specific angular momentum of a satellite about the central attracting body *remains constant*. As r and v change along the orbit, the flight-path angle, ϕ , must change so as to keep h constant. (See Figure 1.4-1 and equation (1.4-4))

1.5.3 Geometrical Properties Common to All Conic Sections. Although Figure 1.5-1 illustrates an ellipse, the ellipse is only one of the family of curves called conic sections. Before discussing the factors

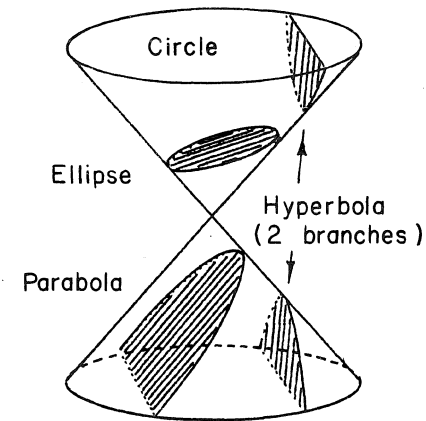


Figure 1.5-2 The conic sections

which determine which of these conic curves a satellite will follow, we need to know a few facts about conic sections in general.

The conic sections have been known and studied for centuries. Many of their most interesting properties were discovered by the early Greeks. The name derives from the fact that a conic section may be defined as the curve of intersection of a plane and a right circular cone. Figure 1.5-2 illustrates this definition. If the plane cuts across one nappe (half-cone), the section is an *ellipse*. A *circle* is just a special case of the ellipse where the plane is parallel to the base of the cone. If, in addition to cutting just one nappe of the cone, the plane is parallel to a line in the surface of the cone, the section is a *parabola*. If the plane cuts both nappes, the section is a *hyperbola* having two branches. There are also *degenerate conics* consisting of one or two straight lines, or a single point, these are produced by planes passing through the apex of the cone.

There is an alternate definition of a conic which is mathematically equivalent to the geometrical definition above:

A conic is a circle or the locus of a point which moves so that the ratio of its absolute distance from a given point (a focus) to

its absolute distance from a given line (a directrix) is a positive constant e (the eccentricity).

While the directrix has no physical significance as far as orbits are concerned, the focus and eccentricity are indispensable concepts in the understanding of orbital motion. Figure 1.5-3 below illustrates certain other geometrical dimensions and relationships which are common to all conic sections.

Because of their symmetry all conic sections have two foci, F and F' . The prime focus, F , marks the location of the central attracting body in an orbit. The secondary or vacant focus, F' , has little significance in

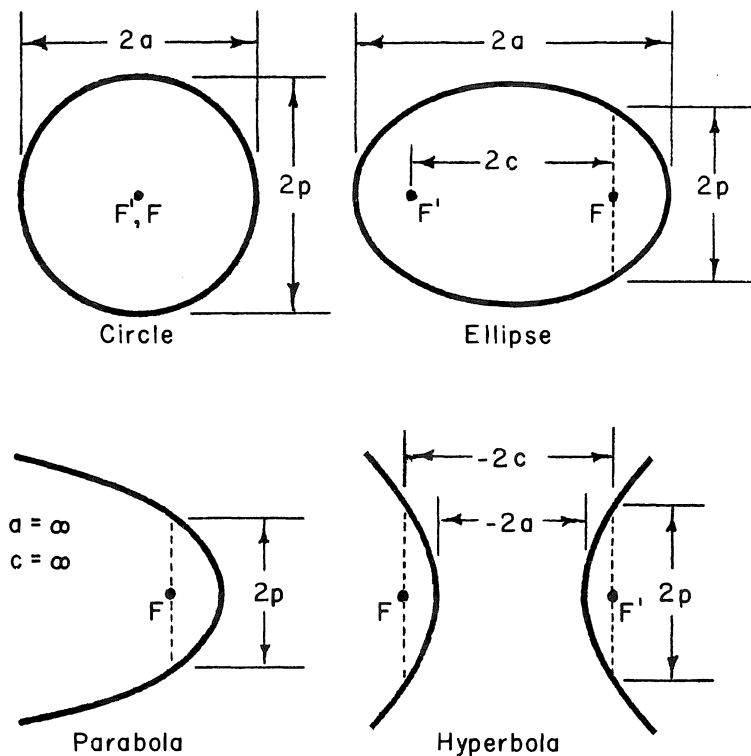


Figure 1.5-3 Geometrical dimensions common to all conic sections

orbital mechanics. In the parabola, which represents the borderline case between the open and closed orbits, the secondary focus is assumed to lie an infinite distance to the left of F . The width of each curve *at the focus* is a positive dimension called the latus rectum and is labeled $2p$ in Figure 1.5-3. The length of the chord passing through the foci is called the major axis of the conic and is labeled $2a$. The dimension, a , is called the semi-major axis. Note that for the circle $2a$ is simply the diameter, for the parabola $2a$ is infinite and for the hyperbola $2a$ is taken as negative. The distance between the foci is given the symbol $2c$. For the circle the foci are considered coincident and $2c$ is zero, for the parabola $2c$ is infinite and for the hyperbola $2c$ is taken as a negative. It follows directly from the definition of a conic section given above that for any conic *except a parabola*

$$e = \frac{c}{a} \quad (1.5-5)$$

and

$$p = a(1 - e^2). \quad (1.5-6)$$

The extreme end-points of the major axis of an orbit are referred to as "apses". The point nearest the prime focus is called "periapsis" (meaning the "near apse") and the point farthest from the prime focus is called "apoapsis" (meaning the "far apse"). Depending on what is the central attracting body in an orbital situation these points may also be called "perigee" or "apogee," "perihelion" or "aphelion," "periselenium" or "aposelenium," etc. Notice that for the circle these points are not uniquely defined and for the open curves (parabolas and hyperbolas) the apoapsis has no physical meaning.

The distance from the prime focus to either periapsis or apoapsis (where it exists) can be expressed by simply inserting $\nu = 0^\circ$ or $\nu = 180^\circ$ in the general polar equation of a conic section (equation (2.5-4)). Thus, *for any conic*

$$r_{\min} = r_{\text{periapsis}} = \frac{p}{1 + e \cos 0^\circ}.$$

Combining this with equation (1.5-6) gives

$$r_p = \frac{p}{1+e} = a(1-e). \quad (1.5-7)$$

Similarly

$$r_{\max} = r_{\text{apoapsis}} = \frac{p}{1+e \cos 180^\circ}$$

and

$$r_a = \frac{p}{1-e} = a(1+e). \quad (1.5-8)$$

1.5.4 The Eccentricity Vector. In the derivation of equation (1.5-3), the trajectory equation, we encountered the vector constant of integration, $\mathbf{B}$, which points toward periapsis. By comparing equations (1.5-3) and (1.5-4) we conclude that $\mathbf{B} = \mu \mathbf{e}$. Quite obviously, since $\mathbf{e}$ is also a constant vector pointing toward periapsis.,

$$\mathbf{e} = \mathbf{B}/\mu. \quad (1.5-9)$$

By integrating the two-body equation of motion we obtained the following result

$$\mathbf{r} \times \mathbf{h} = \mu \frac{\mathbf{r}}{r} + \mathbf{B} \quad (1.5-2)$$

Solving for $\mathbf{B}$ and noting that

$$\mathbf{B} = \mathbf{v} \times \mathbf{h} - \mu \frac{\mathbf{r}}{r}.$$

Hence

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}. \quad (1.5-10)$$

We can eliminate $\mathbf{h}$ from this expression by substituting $\mathbf{h} = \mathbf{r} \times \mathbf{v}$, so

$$\mu \mathbf{e} = \mathbf{v} \times (\mathbf{r} \times \mathbf{v}) - \mu \frac{\mathbf{r}}{r}.$$

Expanding the vector triple product, we get

$$\mu \mathbf{e} = (\mathbf{v} \cdot \mathbf{v}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v} - \mu \frac{\mathbf{r}}{r}.$$

Noting that $(\mathbf{v} \cdot \mathbf{v}) = v^2$ and collecting terms,

$$\mu \mathbf{e} = (v^2 - \frac{\mu}{r}) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v}. \quad (1.5-11)$$

The eccentricity vector will be used in orbit determination in Chapter 2.

1.6 RELATING $\mathbf{e}$ AND $\mathbf{h}$ TO THE GEOMETRY OF AN ORBIT

By comparing equation (1.5-3) and equation (1.5-4) we see immediately that the parameter or semi-latus rectum, p , of the orbit depends only on the specific angular momentum, h , of the satellite. By inspection, *for any orbit*,

$$p = h^2/\mu. \quad (1.6-1)$$

In order to see intuitively why an increase in h should result in a larger value for p consider the following argument:

Suppose that a cannon were set up on the top of a high mountain whose summit extends above the sensible atmosphere (so that we may neglect atmospheric drag). If the muzzle of the cannon is aimed horizontally and the cannon is fired, equation (1.4-4) tells us that $h = rv$ since the flight-path angle, ϕ , is zero. Therefore, progressively increasing the muzzle velocity, v , is equivalent to increasing h . Figure 1.6-1 shows the family of curves which represent the trajectory or orbit of the cannonball as the angular momentum of the "cannonball satellite" is progressively increased. Notice that each trajectory is a conic section with the focus located at the center of the earth, and that as h is

increased the parameter, p , of the orbit also increases just as equation (1.6-1) predicts.

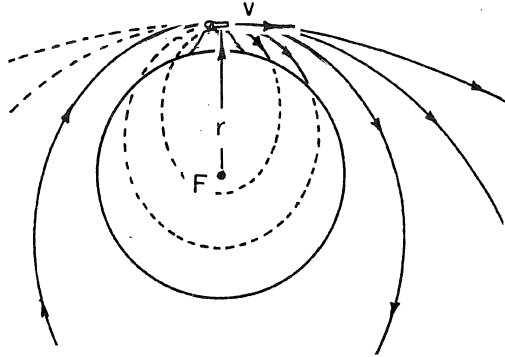


Figure 1.6-1 "Cannonball satellite"

As a by-product of this example we note that at periapsis or apoapsis of any conic orbit the velocity vector (which is always tangent to the orbit) is directed horizontally and the flight-path angle, ν , is zero. We can then write, as a corollary to equation (1.4-4),

$$h = r_p v_p = r_a v_a. \quad (1.6-2)$$

If we write the energy equation (1.4-2) for the periapsis point and substitute from equation (1.6-2) we obtain

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r} = \frac{h^2}{2r_p^2} - \frac{\mu}{r_p}.$$

But from equation (1.5-7)

$$r_p = a(1 - e)$$

and from equation (1.5-6) and equation (1.6-1)

$$h^2 = \mu a(1 - e^2),$$

therefore

$$\mathcal{E} = \frac{\mu a(1 - e^2)}{2a^2(1 - e)^2} - \frac{\mu}{a(1 - e)}$$

which reduces to

$$\mathcal{E} = -\frac{\mu}{2a}. \quad (1.6-3)$$

This simple relationship which is valid *for all conic orbits* tells us that the semi-major axis, a , of an orbit depends only on the specific mechanical energy, $\mathcal{E}$, of the satellite (which in turn depends only on r and v at any point along the orbit). Figure 1.6-1 serves as well to illustrate the intuitive explanation of this fundamental relationship since progressively increasing the muzzle velocity of the cannon also progressively increases $\mathcal{E}$.

Many students find equation (1.6-3) misleading. You should study it together with Figure 1.5-3 which shows that for the circle and ellipse a is positive, while for the parabola a is infinite and for the hyperbola a is negative. This implies that the specific mechanical energy of a satellite in a closed orbit (circle or ellipse) is negative, while $\mathcal{E}$ for a satellite in a parabolic orbit is zero and on a hyperbolic orbit the energy is positive. Thus, the energy of a satellite (negative, zero, or positive) alone determines the type of conic orbit the satellite is in.

Since h alone determines p and since $\mathcal{E}$ alone determines a , the two together determine e (which specifies the exact shape of conic orbit). This can be shown as follows:

$$p = a(1 - e^2) \quad \text{therefore} \quad e = \sqrt{1 - \frac{p}{a}}$$

$$p = h^2/\mu \quad \text{and} \quad a = -\mu/2\mathcal{E}$$

so, for any conic orbit,

$$e = \sqrt{1 + \frac{2\mathcal{E}h^2}{\mu^2}} \quad (1.6-4)$$

Notice again that if $\mathcal{E}$ is negative, e is positive and less than 1 (an ellipse); if $\mathcal{E}$ is zero, e is exactly 1 (a parabola); if $\mathcal{E}$ is positive, e is greater than 1 (a hyperbola). But what if h is zero regardless of what value $\mathcal{E}$ has? The eccentricity will be exactly 1 but the orbit will not be a parabola! Rather, the orbit will be a degenerate conic (a point or straight line). The student should be aware of this pitfall. Namely, all parabolas have an eccentricity of 1 but an orbit whose eccentricity is 1 need not be a parabola—it could be a degenerate conic.

EXAMPLE PROBLEM. For a given satellite, $\mathcal{E} = -2.0 \times 10^8 \text{ ft}^2/\text{sec}^2$ and $e = 0.2$. Determine its specific angular momentum, semi-latus rectum, and semi-major axis.

$$a = -\frac{\mu}{2\mathcal{E}} = \underline{3.5198 \times 10^7 \text{ ft}}$$

$$p = a(1 - e^2) = \underline{3.3790 \times 10^7 \text{ ft}}$$

$$h = \sqrt{p\mu} = \underline{6.897 \times 10^{11} \text{ ft}^2/\text{sec}}$$

EXAMPLE PROBLEM. A radar tracking station tells us that a certain decaying weather satellite has $e = 0.1$ and perigee altitude = 200 n. mi. Determine its altitude at apogee, specific mechanical energy, and specific angular momentum.

$$r_p = r_\oplus + 200 = 3643.9 \text{ n.mi.}$$

$$p = r_p (1 + e) = 4008.3 \text{ n.mi.}$$

$$r_a = \frac{p}{1 - e} = 4453.7 \text{ n.mi.}$$

$$\text{altitude at apogee} = r_a - r_\oplus = \underline{1009.8 \text{ n.mi.}}$$

$$= 6.135 \times 10^6 \text{ ft}$$

$$h = \sqrt{p\mu} = \underline{5.855 \times 10^{11} \text{ ft}^2/\text{sec}}$$

$$2a = r_a + r_p = 8097.6 \text{ n.mi.}$$

$$\mathcal{E} = -\frac{\mu}{2a} = \underline{-2.861 \times 10^8 \text{ ft}^2/\text{sec}^2}$$

1.7 THE ELLIPTICAL ORBIT

The orbits of all the planets in the solar system as well as the orbits of all earth satellites are ellipses. Since an ellipse is a closed curve, an object in an elliptical orbit travels the same path over and over. The time for the satellite to go once around its orbit is called the period. We will first look at some geometrical results which apply only to the ellipse and then derive an expression for the period of an elliptical orbit.

1.7.1 Geometry of the Ellipse. An ellipse can be constructed using two pins and a loop of thread. The method is illustrated in Figure 1.7-1. Each pin marks the location of a focus and since the length of the thread is constant, the sum of the distances from any point on an ellipse to each focus ($r + r'$) is a constant. When the pencil is at either end-point of the ellipse it is easy to see that, specifically

$$r + r' = 2a. \quad (1.7-1)$$

By inspection, the radius of periapsis and the radius of apoapsis are related to the major axis of an ellipse as

$$r_p + r_a = 2a. \quad (1.7-2)$$

Also by inspection, the distance between the foci is

$$r_a - r_p = 2c. \quad (1.7-3)$$

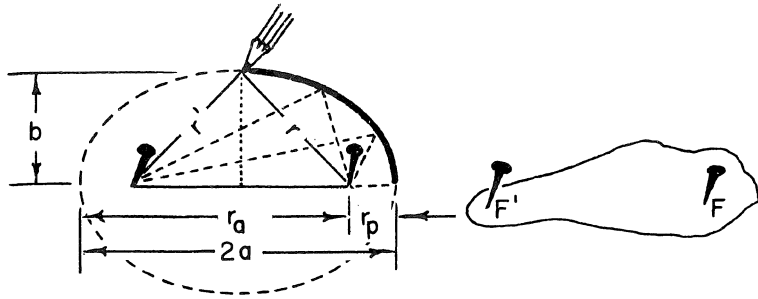


Figure 1.7-1 Simple way to construct an ellipse

Since, in general, e is defined as c/a , equation (1.7-2) and (1.7-3) combine to yield

$$e = \frac{r_a - r_p}{r_a + r_p} \quad (1.7-4)$$

The width of an ellipse at the center is called the minor axis, $2b$. At the end of the minor axis r and r' are equal as illustrated at the right of Figure 1.7-1. Since $r + r' = 2a$, r and r' must both be equal to a at this point. Dropping a perpendicular to the major axis (dotted line in Figure 1.7-1) we can form a right triangle from which we conclude that

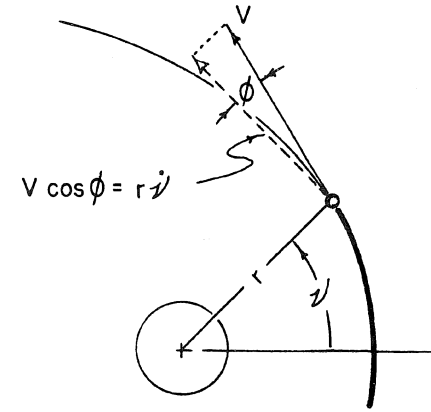
$$a^2 = b^2 + c^2 \quad (1.7-5)$$

1.7.2 Period of an Elliptical Orbit. If you refer to Figure 1.7-2, you will see that the horizontal component of velocity of a satellite is simply $v \cos \phi$ which can also be expressed as $r \dot{\nu}$. Using equation (1.4-4) we can express the specific angular momentum of the satellite as

$$h = \frac{r^2 d\nu}{dt}$$

which, when rearranged, becomes

$$dt = \frac{r^2}{h} d\nu \quad (1.7-6)$$

Figure 1.7-2 Horizontal component of v

But from elementary calculus we know that the differential element of area, dA , swept out by the radius vector as it moves through an angle, $d\nu$, is given by the expression

$$dA = \frac{1}{2} r^2 d\nu$$

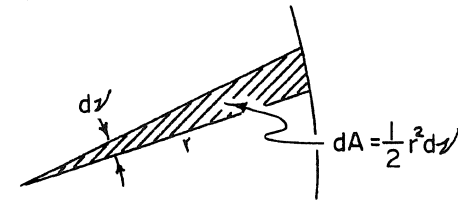


Figure 1.7-3 Differential element of area

So, we can rewrite equation (1.7-6) as

$$dt = \frac{2}{h} dA \quad (1.7-7)$$

Equation (1.7-7) proves Kepler's second law that "equal areas are swept out by the radius vector in equal time intervals" since h is a constant for an orbit.

During one orbital period the radius vector sweeps out the entire area of the ellipse. Integrating equation (1.7-7) for one period gives us

$$\mathcal{T}P = \frac{2\pi ab}{h} \quad (1.7-8)$$

where $\pi a b$ is the total area of an ellipse and $\mathcal{T}P$ is the period. From equations (1.7-5), (1.5-5) and (1.5-6)

$$b = \sqrt{a^2 - c^2} = \sqrt{a^2 (1 - e^2)} = \sqrt{ap}$$

and, since $h = \sqrt{\mu p}$,

$$\boxed{\mathcal{T}P = \frac{2\pi}{\sqrt{\mu}} a^{3/2}} \quad (1.7-9)$$

Thus, the period of an elliptical orbit *depends only on the size of the semi-major axis*, a . Equation (1.7-9), incidentally, proves Kepler's third law that "the square of the period is proportional to the cube of the mean distance" since a , being the average of the periapsis and apoapsis radii, is the "mean distance" of a satellite from the prime focus.

1.8 THE CIRCULAR ORBIT

The circle is just a special case of an ellipse so all the relationships we just derived for the elliptical orbit including the period are also valid for the circular orbit. Of course, the semi-major axis of a circular orbit is just its radius, so equation (1.7-9) is simply

$$\mathcal{T}P_{CS} = \frac{2\pi}{\sqrt{\mu}} r_{CS}^{3/2} \quad (1.8-1)$$

1.8.1 Circular Satellite Speed. The speed necessary to place a satellite in a circular orbit is called circular speed. Naturally, the satellite must be launched in the horizontal direction at the desired

altitude to achieve a circular orbit. The latter condition is called circular velocity and implies both the correct speed and direction. We can calculate the speed required for a circular orbit of radius, r_{CS} from the energy equation.

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

If we remember that $r_{CS} = a$, we obtain

$$\frac{v_{CS}^2}{2} - \frac{\mu}{r_{CS}} = -\frac{\mu}{2r_{CS}}$$

which reduces to

$$\boxed{v_{CS} = \sqrt{\frac{\mu}{r_{CS}}}} \quad (1.8-2)$$

Notice that the greater the radius of the circular orbit the less speed is required to keep the satellite in this orbit. For a low altitude earth orbit, circular speed is about 26,000 ft/sec while the speed required to keep the moon in its orbit around the earth is only about 3,000 ft/sec.

1.9 THE PARABOLIC ORBIT

The parabolic orbit is rarely found in nature although the orbits of some comets approximate a parabola. The parabola is interesting because it represents the borderline case between the open and closed orbits. An object traveling a parabolic path is on a one-way trip to infinity and will never retrace the same path again.

1.9.1 Geometry of the Parabola. There are only a few geometrical properties peculiar to the parabola which you should know. One is that the two arms of a parabola become more and more nearly parallel as one extends them further and further to the left of the focus in Figure 1.9-1. Another is that, since the eccentricity of a parabola is exactly 1, the periapsis radius is just

$$r_p = \frac{p}{2} \quad (1.9-1)$$

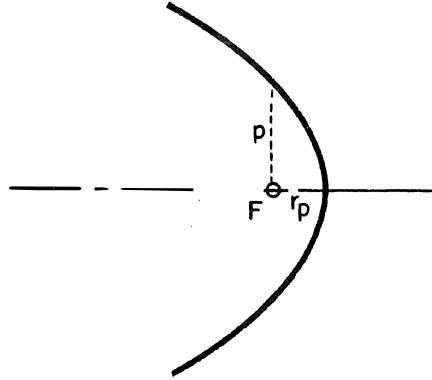


Figure 1.9-1 Geometry of the parabola

which follows from equation (1.5-7). Of course, there is no apoapsis for a parabola and it may be thought of as an “infinitely long ellipse.”

1.9.2 Escape Speed. Even though the gravitational field of the sun or a planet theoretically extends to infinity, its strength decreases so rapidly with distance that only a finite amount of kinetic energy is needed to overcome the effects of gravity and allow an object to coast to an infinite distance without “falling back.” The speed which is just sufficient to do this is called *escape speed*. A space probe which is given escape speed in any direction will travel on a parabolic *escape trajectory*. Theoretically, as its distance from the central body approaches infinity its speed approaches zero. We can calculate the speed necessary to escape by writing the energy equation for two points along the escape trajectory; first at a general point a distance, r , from the center where the “local escape speed” is v_{esc} and then at infinity where the speed will be zero:

$$\mathcal{E} = \frac{v_{esc}^2}{2} - \frac{\mu}{r} = \frac{v_{\infty}^2}{2} - \frac{\mu}{r_{\infty}} = 0$$

from which

$$v_{esc} = \sqrt{\frac{2\mu}{r}}$$

(1.9-2)

Since the specific mechanical energy, $\mathcal{E}$, must be zero if the probe is to have zero speed at infinity and since $\mathcal{E} = -\mu/2a$, the semi-major axis, a , of the escape trajectory must be infinite which confirms that it is a parabola.

As you would expect, the farther away you are from the central body (larger value of r) the less speed it takes to escape the remainder of the gravitational field. Escape speed from the surface of the earth is about 36,700 ft/sec while from a point 3,400 nm above the surface it is only 26,000 ft/sec.

EXAMPLE PROBLEM. A space probe is to be launched on an escape trajectory from a circular parking orbit which is at an altitude of 100 n mi above the earth. Calculate the minimum escape speed required to escape from the parking orbit altitude. (Ignore the gravitational forces of the sun and other planets.) Sketch the escape trajectory and the circular parking orbit.

a. Escape Speed:

Earth gravitational parameter is

$$\mu = 1.407654 \times 10^{16} \text{ ft}^3/\text{sec}^2$$

Radius of circular orbit is

$$\begin{aligned} r &= r_{\text{earth}} + \text{Altitude Circular Orbit} \\ &= 21.53374 \times 10^6 \text{ ft} \end{aligned}$$

From equation (1.9-2)

$$v_{esc} = \sqrt{\frac{2\mu}{r}} = 36,157.9 \text{ ft/sec}$$

b. Sketch of escape trajectory and circular parking orbit:

From the definition of escape speed the energy constant is zero on the escape trajectory which is therefore parabolic. The parameter p is determined by equation (1.5-7).

$$\begin{aligned}
 p &= r_p (1 + e) \\
 &= 21.53374 \times 10^6 \text{ ft} \times 2 \\
 &= 43.06748 \times 10^6 \text{ ft} \\
 &= 7087.8 \text{ n.mi.}
 \end{aligned}$$

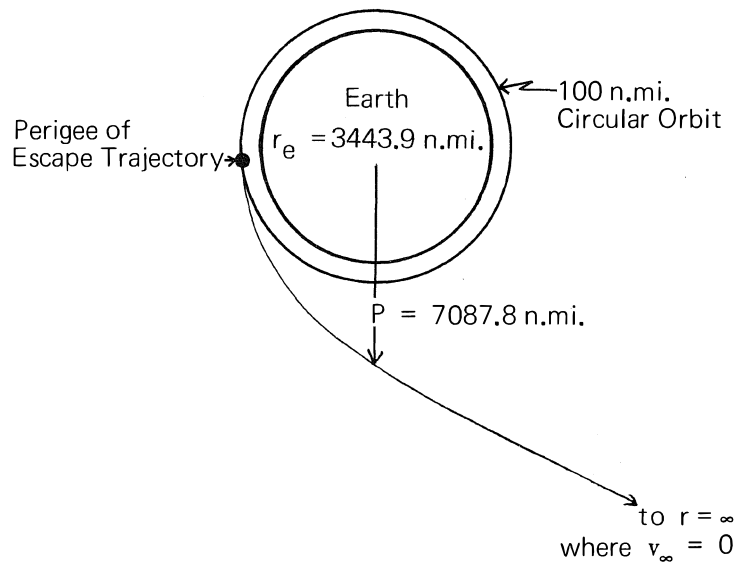


Figure 1.9-2 Escape trajectory for example problem

1.10 THE HYPERBOLIC ORBIT

Meteors which strike the earth and interplanetary probes sent from the earth travel hyperbolic paths relative to the earth. A hyperbolic orbit is necessary if we want the probe to have some speed left over after it escapes the earth's gravitational field. The hyperbola is an unusual and interesting conic section because it has two branches. Its geometry is worth a few moments of study.

1.10.1 Geometry of the Hyperbola. The arms of a hyperbola are asymptotic to two intersecting straight lines (the asymptotes). If we consider the left-hand focus, F , as the prime focus (where the center of

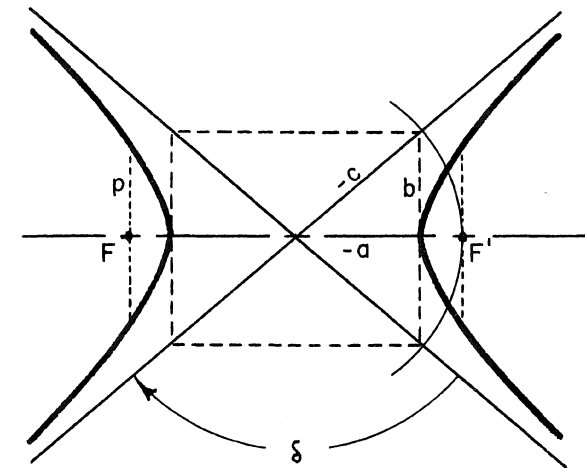


Figure 1.10-1 Geometry of the hyperbola

our gravitating body is located), then only the left branch of the hyperbola represents the possible orbit. If, instead, we assume a force of repulsion between our satellite and the body located at F (such as the force between two like-charged electrical particles), then the right-hand branch represents the orbit. The parameters a , b and c are labeled in Figure 1.10-1. Obviously,

$$c^2 = a^2 + b^2 \quad (1.10-1)$$

for the hyperbola. The angle between the asymptotes, which represents the angle through which the path of a space probe is turned by its encounter with a planet, is labeled δ (delta) in Figure 1.10-1. The turning angle, δ , is related to the geometry of the hyperbola as follows:

$$\sin \frac{\delta}{2} = \frac{a}{c} \quad (1.10-2)$$

but since $e = c/a$ equation (1.10-2) becomes

$$\boxed{\sin \frac{\delta}{2} = \frac{1}{e}} \quad (1.10-3)$$

The greater the eccentricity of the hyperbola, the smaller will be the turning angle, δ .

1.10.2 Hyperbolic Excess Speed. If you give a space probe exactly escape speed, it will just barely escape the gravitational field which

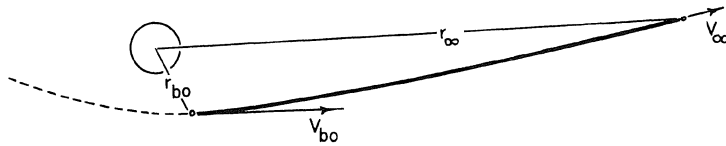


Figure 1.10-2 Hyperbolic excess speed

means that its speed will be approaching zero as its distance from the force center approaches infinity. If, on the other hand, we give our probe more than escape speed at a point near the earth, we would expect the speed at a great distance from the earth to be approaching some finite constant value. This residual speed which the probe would have left over even at infinity is called “hyperbolic excess speed.” We can calculate this speed from the energy equation written for two points on the hyperbolic escape trajectory—a point near the earth called the “burnout point” and a point an infinite distance from the earth where the speed will be the hyperbolic excess speed, v_∞ .

Since specific mechanical energy does not change along an orbit, we may equate $\mathcal{E}$ at the burnout point and $\mathcal{E}$ at infinity:

$$\mathcal{E} = \frac{v_{bo}^2}{2} - \frac{\mu}{r_{bo}} = \frac{v_\infty^2}{2} - \frac{\mu}{r_\infty} \quad (1.10-4)$$

from which we conclude that

$$v_\infty^2 = v_{bo}^2 - \frac{2\mu}{r_{bo}} = v_{bo}^2 - v_{esc}^2 \quad (1.10-5)$$

Note that if v_∞ is zero (as it is on a parabolic trajectory) v_{bo} becomes simply the escape speed.

1.10.3 Sphere of Influence. It is, of course, absurd to talk about a space probe “reaching infinity” and in this sense it is meaningless to talk about escaping a gravitational field completely. It is a fact, however, that once a space probe is a great distance (say, a million miles) from earth, for all practical purposes it has escaped. In other words, it has already slowed down to very nearly its hyperbolic excess speed. It is convenient to define a sphere around every gravitational body and say that when a probe crosses the edge of this “sphere of influence” it has escaped. Although it is difficult to get even two people to agree on exactly where the sphere of influence should be drawn, the concept is convenient and is widely used, especially in lunar and interplanetary trajectories.

1.11 CANONICAL UNITS

Astronomers are as yet unable to determine the precise distance and mass of objects in space. Such fundamental quantities as the mean distance from the earth to the sun, the mass and mean distance of the moon and the mass of the sun are not accurately known. This dilemma is avoided in mathematical calculations if we assume the mass of the sun to be 1 “mass unit” and the mean distance from the earth to the sun to be our unit of distance which is called an “astronomical unit.” All other masses and distances can then be given in terms of these assumed units even though we do not know precisely the absolute value of the sun’s mass and distance in pounds or miles. Astronomers call this

normalized system of units “canonical units.”

We will adopt a similar system of normalized units in this text primarily for the purpose of simplifying the arithmetic of our orbit calculations.

1.11.1 The Reference Orbit. We will use a system of units based on a hypothetical circular reference orbit. In a two-body problem where the sun is the central body the reference orbit will be a circular orbit whose radius is one astronomical unit (AU). For other problems where the earth, moon, or some other planet is the central body the reference orbit will be a minimum altitude circular orbit just grazing the surface of the planet.

We will define our distance unit (DU) to be the radius of the reference orbit. If we now define our time unit (TU) such that the speed of a satellite in the hypothetical reference orbit is 1 DU/TU, then the value of the gravitational parameter, μ , will turn out to be 1 DU³/TU².

Unless it is perfectly clear which reference orbit the units in your problem are based on you will have to indicate this by means of a subscript on the symbol DU and TU. This is most easily done by annexing as a subscript the astronomer’s symbol for the sun, earth, or other planet. The most commonly used symbols are:

☉ The Sun	♃ Jupiter
☾ The Moon	♄ Saturn
☿ Mercury	♅ Uranus
♀ Venus	♆ Neptune
♁ The Earth	♇ Pluto
♂ Mars	

The concept of the reference orbit is illustrated in Figure 1.11-1.

Values for the commonly used astrodynamic constants and their relationship to canonical units are listed in the appendices.

EXAMPLE PROBLEM. A space object is sighted at an altitude of 1.046284×10^7 ft above the earth traveling at 2.593625×10^4 ft/sec and a flight path angle of 0° at the time of sighting. Using canonical units determine $\mathcal{E}$, h , p , e , r_a , r_p .

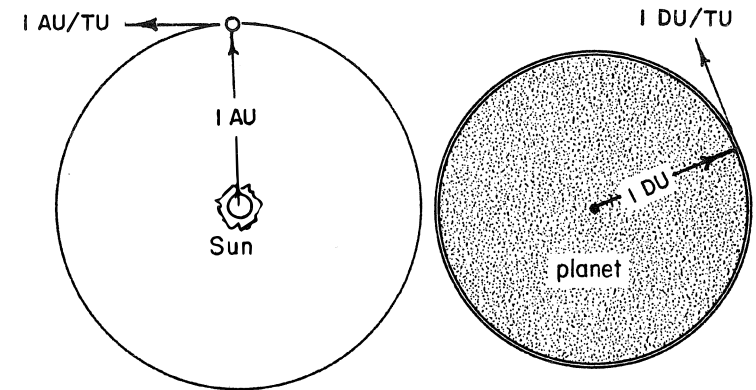


Figure 1.11-1 Reference circular orbits

Convert altitude and speed to earth canonical units.

$$\text{Alt} = .5 \text{ DU}_{\oplus}$$

$$v = 1 \text{ DU}_{\oplus} / \text{TU}_{\oplus}$$

The gravitational parameter and earth radius are:

$$\mu_{\oplus} = 1.407647 \times 10^{16} \text{ ft}^3/\text{sec}^2 = 1 \text{ DU}_{\oplus}^3 / \text{TU}_{\oplus}^2$$

$$r_{\oplus} = 1 \text{ DU}_{\oplus}$$

The radius of the object from the center of the earth is:

$$r = r_{\oplus} + \text{Alt} = 1.5 \text{ DU}_{\oplus}$$

Find $\mathcal{E}$ from equation (1.4-2).

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu_{\oplus}}{r} = -.167 \text{ DU}_{\oplus}^2 / \text{TU}_{\oplus}^2 = -1.12339 \times 10^8 \text{ ft}^2/\text{sec}^2$$

Find h from equation (1.4-4)

$$h = rv \cos \phi = 1.5DU_{\oplus}^2/TU_{\oplus} = 8.141 \times 10^{11} \text{ ft}^2/\text{sec}$$

Find p from equation (1.6-1)

$$p = \frac{h^2}{\mu_{\oplus}} = 2.25DU_{\oplus} = 4.7082763 \times 10^7 \text{ ft}$$

Find e from equation (1.6-4)

$$e = \sqrt{1 + \frac{2\mathcal{E}h^2}{\mu^2}} = .5$$

Find r_a from equation (1.5-8)

$$r_a = \frac{p}{1 - e} = 4.5DU_{\oplus} = 9.416553 \times 10^7 \text{ ft}$$

Find r_p from equation (1.5-7)

$$r_p = \frac{p}{1 + e} = 1.5DU_{\oplus} = 3.138851 \times 10^7 \text{ ft}$$

EXERCISES

1.1 The position and velocity of a satellite at a given instant are described by

$$\mathbf{r} = 2\mathbf{I} + 2\mathbf{J} + 2\mathbf{K} \quad (\text{Distance Units})$$

$$\mathbf{v} = -.4\mathbf{I} + .2\mathbf{J} + .4\mathbf{K} \quad (\text{Distance Units per Time Unit})$$

where $\mathbf{I} \mathbf{J} \mathbf{K}$ is a nonrotating geocentric coordinate system. Find the specific angular momentum and total specific mechanical energy of the satellite.

(Answer: $\mathbf{h} = .4\mathbf{I} - 1.6\mathbf{J} + 1.2\mathbf{K} \text{ DU}^2/\text{TU}$,
 $\mathcal{E} = -.1087 \text{ DU}^2/\text{TU}^2$)

1.2 For a certain satellite the observed velocity and radius at $\nu = 90^\circ$ is observed to be 45,000 ft/sec and 4,000 n mi, respectively. Find the eccentricity of the orbit.
 (Answer: $e = 1.581$)

1.3 An earth satellite is observed to have a height of perigee of 100 n mi and a height of apogee of 600 n mi. Find the period of the orbit.

1.4 Six constants of integration (or effectively, 6 orbital elements) are required for a complete solution to the two-body problem. Why, in general, is a completely determined closed solution of the N-body problem an impossibility if $N \geq 3$?

1.5 For a certain earth satellite it is known that the semi-major axis, a , is $30 \times 10^6 \text{ ft}$. The orbit eccentricity is 0.2.

- Find its perigee and apogee distances from the center of the earth.
- Find the specific energy of the trajectory.
- Find the semi-latus rectum or parameter (p) of the orbit.

d. Find the length of the position vector at a true anomaly of 135° .

(Ans. $r = 3.354 \times 10^7$ ft)

1.6 Find an equation for the velocity of a satellite as a function of total specific mechanical energy and distance from the center of the earth.

1.7 Prove that $r_{\text{apoapsis}} = a(1 + e)$.

1.8 Identify each of the following trajectories as either circular, elliptical, hyperbolic, or parabolic:

a. $r = 3 \text{ DU}$

d. $\mathbf{r} = \mathbf{J} + .2\mathbf{K}$

$v = 1.5 \text{ DU/TU}$

$\mathbf{v} = .9\mathbf{I} + .123\mathbf{K}$

b. $r_{\text{perigee}} = 1.5 \text{ DU}$

e. $r = 1.01\mathbf{K}$

$p = 3 \text{ DU}$

$\mathbf{v} = \mathbf{I} + 1.4\mathbf{K}$

c. $\mathcal{E} = -1/3 \text{ DU}^2/\text{TU}^2$

$p = 1.5 \text{ DU}$

1.9 A space vehicle enters the sensible atmosphere of the earth (300,000 ft) with a velocity of 25,000 ft/sec at a flight-path angle of -60° . What was its velocity and flight-path angle at an altitude of 100n. mi during descent?

(Ans. $v = 24,618 \text{ ft/sec}$, $\phi = -59^\circ 58'$)

1.10 Show that two-body motion is confined to a plane fixed in space.

1.11 A sounding rocket is fired vertically. It achieves a burnout speed of 10,000 ft/sec at an altitude of 100,000 ft. Determine the maximum altitude attained. (Neglect atmospheric drag.)

1.12 Given that $e = \frac{c}{a}$, derive values for e for circles, ellipses and hyperbolas.

1.13 Show by means of the differential calculus that the position vector is an extremum (maximum or minimum) at the apses of the orbit.

1.14 Given the equation $r = \frac{p}{1 + e \cos \nu}$ plot at least four points, sketch and identify the locus and label the major dimensions for the following conic sections:

a. $p = 2$, $e = 0$

b. $p = 6$, $e = .2$

c. $p = 6$, $e = .6$

d. $p = 3$, $e = 1$

e. $p = 2$, $e = 2$

(HINT: Polar graph paper would be of help here!)

1.15 Starting with:

$$m_k \ddot{\mathbf{r}}_k = \sum_{\substack{j=1 \\ j \neq k}}^m \frac{Gm_j m_k}{r_{jk}^2} \frac{\mathbf{r}_j - \mathbf{r}_k}{r_{jk}}$$

where $\mathbf{r}_j$ is the vector from the origin of an inertial frame to any j^{th}

body and r_{jk} is the scalar distance between the j^{th} and k^{th} bodies
 ($r_{jk} \equiv r_{kj}$)

a. Show that:

$$\sum_{k=1}^m m_k \ddot{\mathbf{r}}_k = 0$$

b. Using the definition of a system's mass center show:

$$\mathbf{r}_C = \mathbf{a}t + \mathbf{b}$$

where $\mathbf{r}_C$ is the vector from the inertial origin to the system mass center and $\mathbf{a}$ and $\mathbf{b}$ are constant vectors.

c. What is the significance of the equation derived in part b above?

1.16 A satellite is injected into an elliptical orbit with a semi-major axis equal to $4D U_{\oplus}$. When it is precisely at the end of the semi-minor axis it receives an impulsive velocity change just sufficient to place it into an escape trajectory. What was the magnitude of the velocity change?

(Ans. $\Delta v = 0.207 DU_{\oplus}/TU_{\oplus}$)

1.17 Show that the speed of a satellite on an elliptical orbit at either end of the *minor* axis is the same as local circular satellite speed at that point.

1.18 Show that when an object is located at the intersection of the semi-minor axis of an elliptical orbit the eccentricity of the orbit can be expressed as $e = -\cos \nu$.

* **1.19** BMEWS (Ballistic Missile Early Warning System) detects an unidentified object with the following parameters:

altitude = .5 DU

speed = $\sqrt{2/3}$ DU/TU

flight-path angle = 30°

Is it possible that this object is a space probe intended to escape the earth, an earth satellite or a ballistic missile?

* **1.20** Prove that the flight-path angle is equal to 45° when $\nu = 90^\circ$ on all parabolic trajectories.

* **1.21** Given two spherically symmetric bodies of considerable mass, assume that the only force that acts is a repulsive force, proportional to the product of the masses and inversely proportional to the cube of the distance between the masses that acts along the line connecting the centers of the bodies. Assume that Newton's second law holds ($\Sigma \mathbf{F} = m\mathbf{a}$) and derive a differential equation of motion for these bodies.

* **1.22** A space vehicle destined for Mars was first launched into a 100 n mi circular parking orbit.

a. What was speed of vehicle at injection into parking orbit?

The vehicle coasted in orbit for a period of time to allow system checks to be made and then was restarted to increase its velocity 37,600 ft/sec which placed it on an interplanetary trajectory toward Mars.

b. Find e , h and $\mathcal{E}$ relative to the earth for the escape orbit. What kind of orbit is it?

c. Compare the velocity at 1,000,000 n mi from the earth with the hyperbolic excess velocity, v_∞ . Why are the two so nearly alike?

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